
MFEM WORKSHOP · LIGHTNING TALK

WaveHoltz

A general-purpose GPU Helmholtz solver in MFEM

Amit Rotem

PhD, 2026 · Virginia Tech

Will Pazner

Portland State University

Daniel Appelö

Virginia Tech

PROBLEM & DISCRETIZATION

From the Helmholtz equation to a linear system

HELMHOLTZ EQUATION

$$-\nabla \cdot (c^2 \nabla u) - \omega^2 \rho^2 u = f \quad \text{in } \Omega \subset \mathbb{R}^d$$

Variable wave speed $c(x)$ and density $\rho(x)$; impedance (radiation) condition on a boundary portion $\Gamma_R \subset \partial\Omega$.

↓ Galerkin on hexahedra, degree- p tensor-product elements: continuous (H^1) or SIPDG

BILINEAR FORM

Find u in the finite element space V_h such that, for all $\phi \in V_h$,

$$a(u, \phi) - i\omega \gamma(u, \phi) - \omega^2 m(u, \phi) = b(\phi)$$

$$a = \int_{\Omega} \nabla \bar{\phi} \cdot c^2 \nabla u \, dx, \quad \gamma = \int_{\Gamma_R} \rho u \bar{\phi} \, ds, \quad m = \int_{\Omega} \rho^2 u \bar{\phi} \, dx$$

b is the load functional from f . SIPDG replaces a by a_{DG} , adding interior-penalty face terms.

↓ GLL tensor-product basis; operators applied matrix-free

LINEAR SYSTEM

$$(S - i\omega H - \omega^2 M) \mathbf{u} = \mathbf{b}$$

S, H, M discretize a, γ, m and are symmetric positive semi-definite, so the system is symmetric but **indefinite**; complex and non-Hermitian when $H \neq 0$.

WHY IT IS HARD

- **Indefinite.** Eigenvalues of $S - \omega^2 M$ straddle the origin. Krylov, multigrid and Schwarz methods that excel on definite problems deteriorate or fail; MINRES avoids breakdown but stagnates from ill-conditioning.
- **Resolution.** Dispersion (pollution) error requires $h^p \omega^{p+1} \approx \text{const}$, far beyond $h\omega$ small — motivating high order.
- **Scale.** Realistic 3D problems reach 10^8+ DOFs: the solver must be matrix-free, GPU-ready and $O(n)$ in the number of DOFs n .

THE METHOD

WaveHoltz: Helmholtz via filtered wave propagation

TIME-HARMONIC SOLUTION $\Leftrightarrow$ WAVE EQUATION

If $\mathbf{u}$ solves the Helmholtz system, then $\mathbf{p}(t) = \Re\{\mathbf{u}e^{-i\omega t}\}$ solves the wave equation:

$$\dot{\mathbf{p}} = \mathbf{q}, \quad M\dot{\mathbf{q}} + H\mathbf{q} + S\mathbf{p} = \Re\{\mathbf{b}e^{-i\omega t}\}$$

with $\mathbf{p}(0) = \Re\{\mathbf{u}\}$, $\mathbf{q}(0) = \omega\Im\{\mathbf{u}\}$. WaveHoltz exploits this relationship: evolve a guess $\mathbf{u}^{(k)}$ for P periods and **filter in time** to project onto the ω -harmonic component:

$$\mathbf{u}^{(k+1)} = \Pi\mathbf{u}^{(k)} = \frac{2}{T} \int_0^T \kappa(t) \left[\mathbf{p}(t) - \frac{1}{i\omega} \mathbf{q}(t) \right] dt$$

$$\kappa(t) = \cos \omega t - \frac{1}{4}, \quad T = 2\pi P/\omega$$

DISCRETE ITERATION & KRYLOV ACCELERATION

Discretizing in time with N steps:

$$\Pi\mathbf{u} = \frac{2}{N} \sum_{n=0}^{N-1} \kappa_n \left(\mathbf{p}^n - \frac{1}{i\omega} \mathbf{q}^n \right)$$

$$\kappa_n = \cos(n\omega\Delta t) - \frac{1}{4} + \frac{1}{4} \tan^2 \frac{\omega\Delta t}{2}$$

The shift makes it contractive. Π is affine, $\Pi\mathbf{u} = \mathcal{S}\mathbf{u} + \mathcal{G}\mathbf{b}$ ($\mathcal{S}, \mathcal{G}$ linear), so its fixed point solves

$$(I - \mathcal{S})\mathbf{u} = \mathcal{G}\mathbf{b}$$

- Solved with **GMRES**; contractivity $\Rightarrow$ convergence for **any restart length**, Krylov memory $O(n)$.
- Energy-conserving case (no impedance boundary, $H = 0$): $\mathbf{u}$ real and $M(I - \mathcal{S})$ is SPD.
- Each matvec is P periods of time stepping: **matrix-free, no factorization**, no problem-specific tuning.
- Tuned in practice: $P = 2$ periods, about 5 steps per period.

THE METHOD

ω -exact time integrators

Both schemes are **ω -exact**: they evolve $e^{-i\omega t}$ exactly, so the fixed point $\mathbf{u}^{(k)} \rightarrow \mathbf{u}$ is the solution of the spatially discrete Helmholtz system, with **no time-stepping error**; Δt affects only cost and stability.

EXPLICIT – DG

$$(M + \theta H) \mathbf{q}^{n+\frac{1}{2}} = (M - \theta H) \mathbf{q}^{n-\frac{1}{2}} - \sigma S \mathbf{p}^n + \sigma \Re\{\mathbf{b} e^{-in\omega\Delta t}\},$$

$$\mathbf{p}^{n+1} = \mathbf{p}^n + \sigma \mathbf{q}^{n+\frac{1}{2}}$$

- $M + \theta H$ is **block-diagonal** for DG: independent element-local SPD solves.
- No global solve; conditionally stable.

IMPLICIT – CG OR DG: BLOCK SYSTEM

$$\begin{pmatrix} M & -\theta M \\ \theta S & M + \theta H \end{pmatrix} \begin{pmatrix} \mathbf{p}^{n+1} \\ \mathbf{q}^{n+1} \end{pmatrix} = \begin{pmatrix} M & \theta M \\ -\theta S & M - \theta H \end{pmatrix} \begin{pmatrix} \mathbf{p}^n \\ \mathbf{q}^n \end{pmatrix} + \sigma \Re\left\{e^{-i(n+\frac{1}{2})\omega\Delta t} \begin{pmatrix} 0 \\ \mathbf{b} \end{pmatrix}\right\}$$

Schur complement, with $\mathbf{q}^{n+1} = \mathbf{q}^n + 2\theta\mathbf{z}$:

$$(\theta^2 S + \theta H + M) \mathbf{z} = -S(\mathbf{p}^n + \theta \mathbf{q}^n) - H \mathbf{q}^n + \frac{\sigma}{2\theta} \Re\{e^{-i(n+\frac{1}{2})\omega\Delta t} \mathbf{b}\},$$

$$\mathbf{p}^{n+1} = \mathbf{p}^n + \theta (\mathbf{q}^n + \mathbf{q}^{n+1})$$

- $\theta^2 S + \theta H + M$ is **SPD**; stable for $\omega\Delta t < \pi$, independent of h .
- Solved by **CG + LOR-AMG** (next slide).

$$\Delta t = \frac{2\pi P}{\omega N}, \quad \theta = \frac{\tan(\omega\Delta t/2)}{\omega}, \quad \sigma = \frac{\sin(\omega\Delta t/2)}{\omega/2}$$

IMPLEMENTATION

Mapping the algorithm onto MFEM

OBJECT	MFEM COMPONENT
M, S, H	<code>ParBilinearForm</code> with <code>AssemblyLevel::PARTIAL</code> : matrix-free, sum-factorized kernels on E-vectors; vectors on device (CUDA / HIP).
M mass	<code>MassIntegrator</code> (ρ^2)
S diffusion	<code>DiffusionIntegrator</code> (c^2); DG adds <code>DGDiffusionIntegrator</code> (SIPG, penalty κ) on interior and Dirichlet faces — matrix-free face kernels (next slide).
H impedance	CG: <code>MassIntegrator</code> (ρ) on the impedance boundary. DG: <code>BoundaryMassIntegrator</code> (ρ) on impedance faces.
Explicit $(M + \theta H)^{-1}$	<code>CGSolver</code> preconditioned by <code>DGMassInverse</code> (block-diagonal DG mass: element-local, matrix-free, fused kernel); typically 2–3 CG iterations . Future: one fused kernel for $(M + \theta H)^{-1}$, like <code>DGMassInverse</code> .
Implicit $\theta^2 S + \theta H + M$	Another PA <code>ParBilinearForm</code> (coefficients $\theta^2 c^2$, $\theta \rho$, ρ^2); <code>CGSolver</code> + <code>LORSolver</code> < <code>HypreBoomerAMG</code> >.
Outer loop	A <code>WaveHoltz : Operator</code> whose <code>Mult</code> is evolve-and-filter; <code>GMRESSolver</code> with <code>SetKDim</code> .

LOR-AMG FOR THE IMPLICIT SOLVE

AMG on the assembled high-order matrix degrades with p : each element's $(p+1)^d$ DOFs are fully coupled, so the stencil grows dense. Instead, `LORSolver` builds the sparse **low-order-refined** operator on the GLL sub-mesh (one DOF per DOF) and applies `HypreBoomerAMG`:

$$\tilde{\mathcal{A}} = \theta^2 \tilde{S} + \theta \tilde{H} + \tilde{M}$$

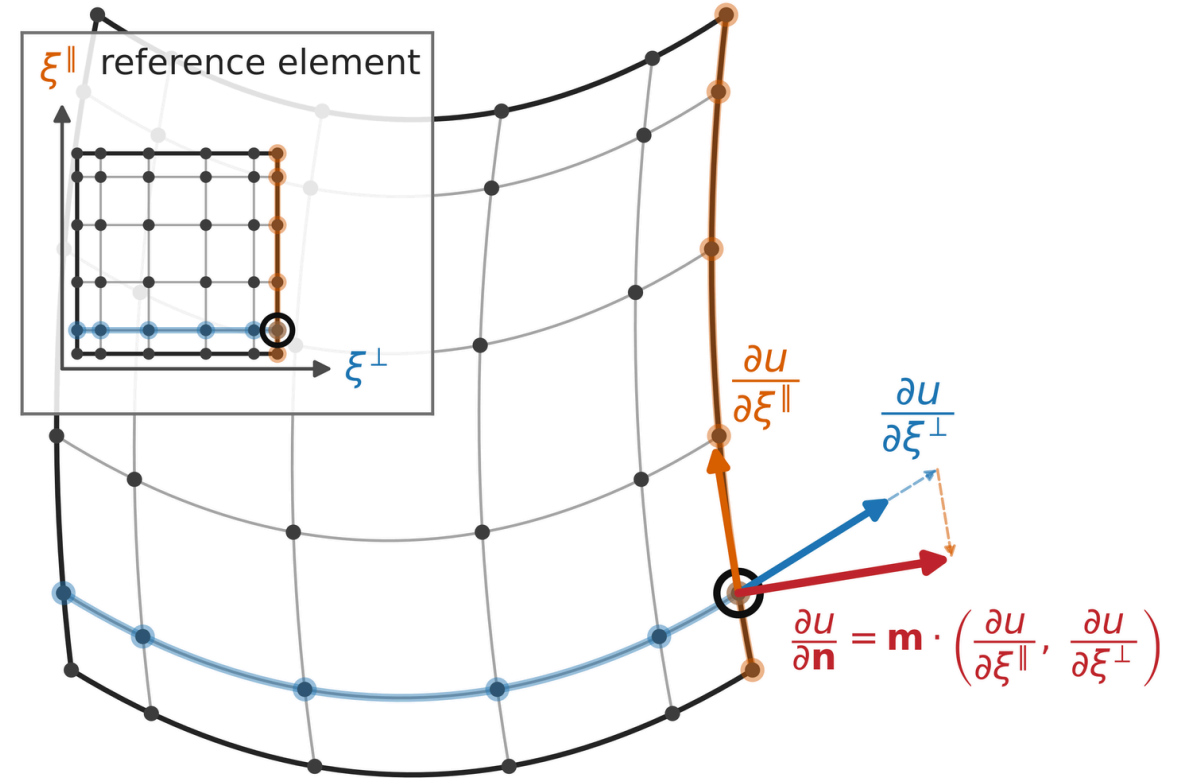
- Spectrally equivalent to $\mathcal{A} = \theta^2 S + \theta H + M$, with constants **independent of p and h** $\Rightarrow$ bounded CG iterations.
- Setup and V-cycle cost $O(n)$, independent of p : total solve cost linear in DOFs.

Matrix-free DG diffusion face kernels

- Volume terms act on **E-vectors**; face terms act on **F-vectors** (face values from both sides of each face).
- SIPDG needs $\partial_{\mathbf{n}} u = \mathbf{m} \cdot \nabla_{\xi} u$, $\mathbf{m} = J^{-1} \mathbf{n}$ (J : element Jacobian). Evaluating it at high-degree quadrature would inflate memory traffic.
- Instead store only the **reference-normal derivative** $\partial u / \partial \xi^{\perp}$ (degree p); the tangential derivative is a 1D contraction of face values already resident.

$$\frac{\partial u}{\partial \mathbf{n}} = \mathbf{m} \cdot \left(\frac{\partial u}{\partial \xi^{\parallel}}, \frac{\partial u}{\partial \xi^{\perp}} \right)$$

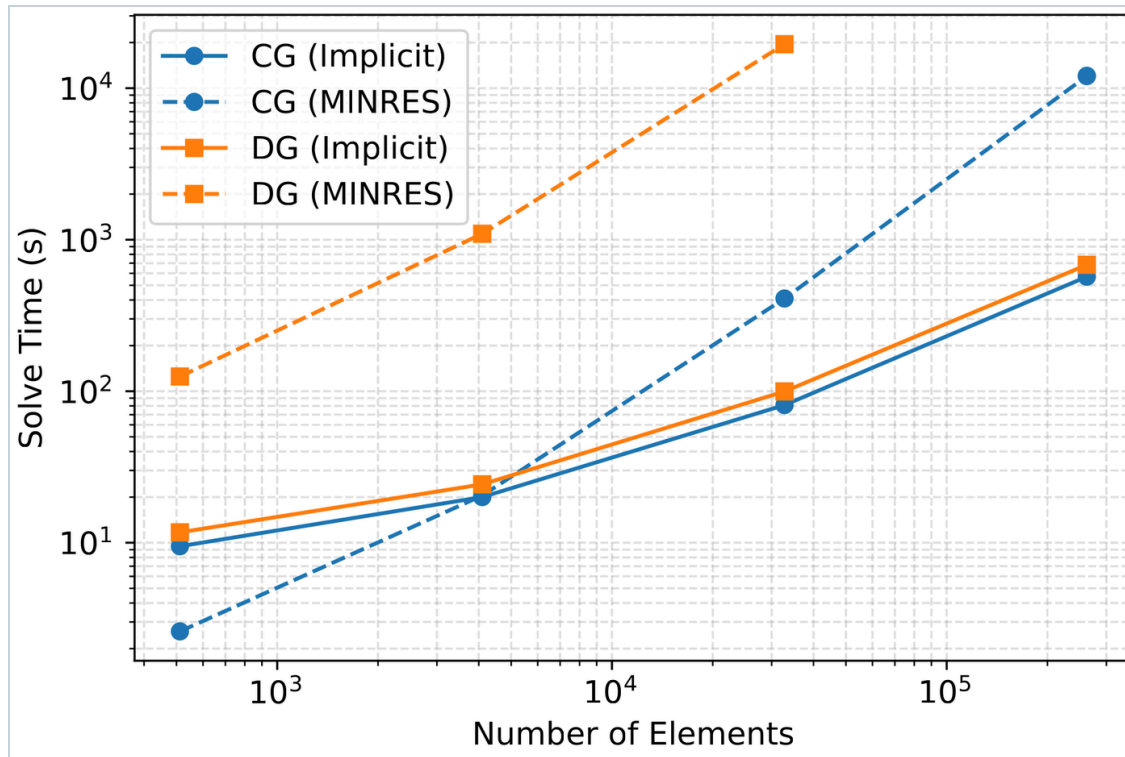
MFEM: `DGDiffusionIntegrator` with
`RequiresFaceNormalDerivatives()` and
`AddMultPAFaceNormalDerivatives`.



Normal derivative at a face DOF of a curved 2D element: reference-normal (blue) and reference-tangential (orange) derivatives combine into $\partial_{\mathbf{n}} u$ (red).

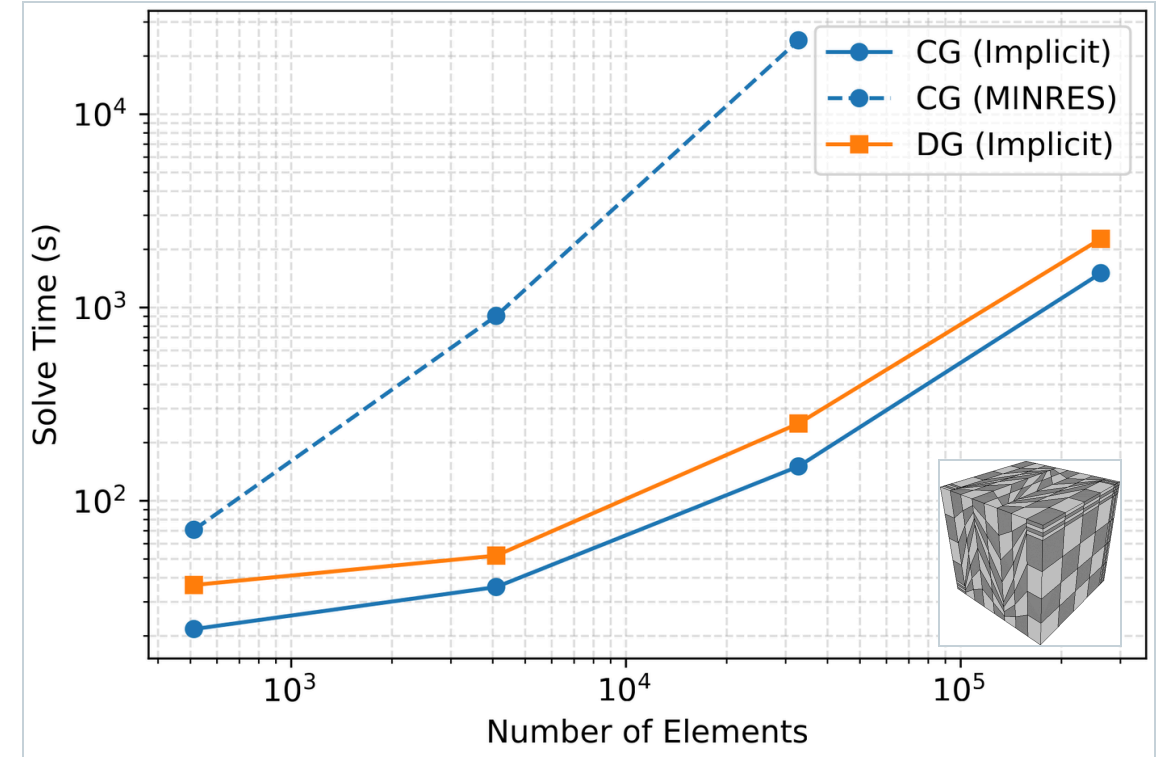
RESULTS • EFFICIENCY & ROBUSTNESS

Against MINRES: efficient, and insensitive to mesh quality

UNIFORM MESH — $\omega = 10$, $p = 4$, ONE A100Solve time to 10^{-6} ; solid: WaveHoltz, dashed: MINRES.

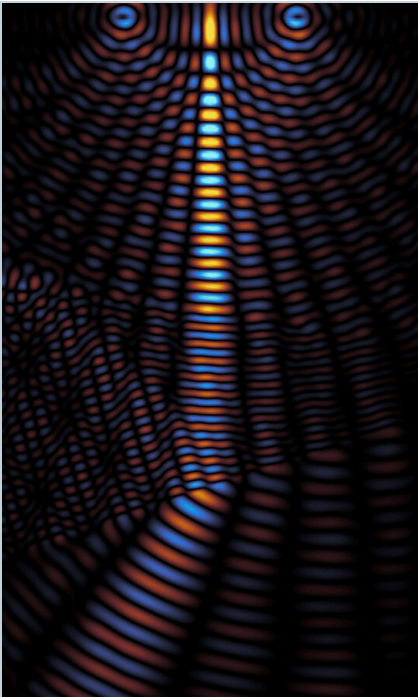
- WaveHoltz iteration count is **effectively constant** in problem size; MINRES grows with it.
- Largest CG problem: **>20× faster**. Largest DG problem: **11 min** vs. not converged in 20 h.

KERSHAW MESH — SEVERE SKEW, ILL-CONDITIONED

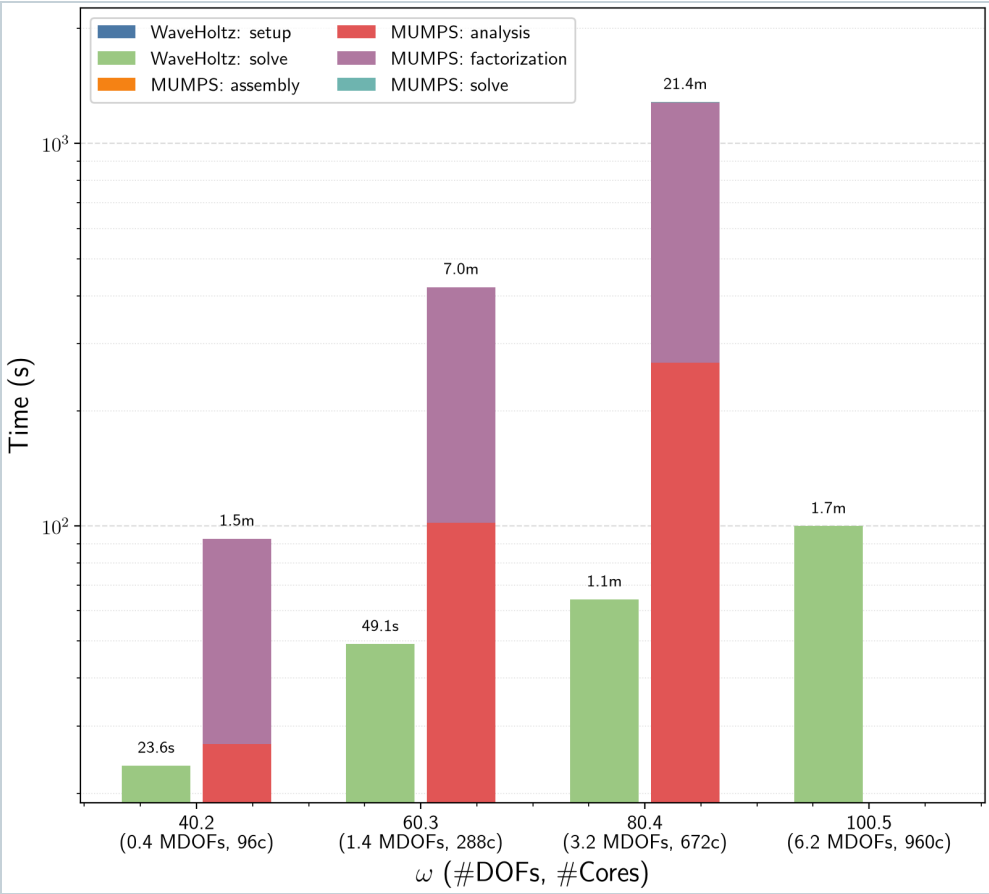
Solve time to 10^{-6} ; solid: WaveHoltz, dashed: MINRES (DG omitted).

- DG MINRES: **5.2M iterations** ($\approx 80\times$ the DOFs) on the smallest mesh. CG MINRES: >24 h on the largest; WaveHoltz: <1 h.
- WaveHoltz iterations: **37** $\rightarrow$ **40** vs. the uniform mesh; $\leq 2.7\times$ (CG), $\leq 3.7\times$ (DG) slower.

Against MUMPS: 10× faster, a fraction of the memory



$\omega = 400, p = 4$: 24.3M DOF on a single A100 in 27 min.



Wall time, $p = 2, \approx 10^4$ DOF/core, $h\omega = 2\pi/10$, 96-core AMD EPYC nodes.

$\text{nnz}(LU) \gtrsim O(n^{3/2}), \quad n = Np^d \text{ (} N \text{ elements)}$
WaveHoltz memory: $O(n)$

ω	DOFS	CORES	WAVEHOLTZ	MUMPS
40.2	0.41M	96	13.8 GB	46.7 GB
60.3	1.36M	288	65.3 GB	267 GB
80.4	3.21M	672	153 GB	940 GB
100.5	6.25M	960	227 GB	OOM

Total memory. At 6.2M DOF MUMPS exceeds 512 GB/node; WaveHoltz needs 23 GB/node.

- WaveHoltz converges before MUMPS finishes its symbolic analysis.
- MUMPS's cheap solve phase wins for many right-hand sides; frequency sweeps and adaptive refinement demand refactorization.



SCALE

541 million degrees of freedom

Acoustic scattering off a submarine hull, $p = 7$, $\omega = 20$. The same code ran on NVIDIA A30 GPUs (**ORCA**, Portland State) and AMD MI300A APUs (**Tuolumne**, LLNL); on the 68M-DOF version the MI300A run was 6× faster.

82 min

total solve time

32 GPUs

AMD MI300A, Tuolumne

1.57M

elements, $p = 7$

94%

strong scaling from 16 GPUs

TAKEAWAYS

A general-purpose Helmholtz solver

Robust

Iteration counts are insensitive to mesh quality, frequency and coefficients. The contractive fixed-point structure makes GMRES safe at any restart length.

Efficient

Iterations essentially independent of problem size; linear time and memory. Orders of magnitude faster than MINRES, an order of magnitude faster than MUMPS at a fraction of the memory.

MFEM-native

Partial-assembly operators, `DGMassInverse`, and `LORSolver<HypreBoomerAMG>`, plus two ω -exact time integrators on top.

Scalable & portable

Solves 541M-DOF problems on 16–32 MI300A GPUs with good strong scaling on this problem; the same code runs on NVIDIA and AMD hardware without vendor-specific tuning.

Questions?

Built with MFEM (mfem.org)

Implementation off-the-shelf MFEM partial-assembly operators, plus two new ω -exact time integrators

REFERENCES

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